

sustAInability Project Report

Challenge

Digitalrat Challenge — “How might we connect elderly people with students and AI so that they can help them in the digital space?”

Team & Project

Team BridgeBuilders — BridgeBuilders: a web app connecting student volunteers with senior citizens for guided, in-person digital assistance.

Mission Statement

To help older adults in Munich develop digital skills and confidence, enabling them to participate fully, safely, and independently in an increasingly digital society.

1. Introduction

Digitalization is reshaping public administration, communication and everyday services across Germany and the European Union. Processes that once required a visit to a local office — opening a bank account, registering a residence, or purchasing a train ticket — increasingly take place online, often gated behind digital identity systems such as BundID. While this shift promises efficiency, it also creates a growing gap for citizens who are less digitally literate, most notably senior citizens.

Our team, BridgeBuilders, was formed to address the Digitalrat Challenge: “How might we connect elderly people with students and AI so that they can help them in the digital space?” We approached this challenge through a structured design-thinking process — problem framing, ideation, clustering and prioritization — which converged on a single core idea: a web application that connects student volunteers with senior citizens for face-to-face digital assistance, supported by structured guides and an AI assistant.

This report documents the problem we investigated (Section 2), the solution we designed (Section 3), the guides we created for real assistance scenarios (Section 4), the legal safeguards built into our concept (Section 5), the functional prototype we implemented (Section 6), and a discussion of impact, limitations and next steps (Section 7).

2. Problem Analysis

Public administration in Germany has undergone rapid digital transformation. Since the 2017 Onlinezugangsgesetz (Online Access Act), federal, state and municipal authorities have been obligated to offer administrative services digitally, and the BundID system now serves as the central identity-verification layer for e-government services such as tax filing, residence registration, and benefits applications. For citizens comfortable with digital tools, this shift lowers friction. For others, it introduces a new barrier before they can even begin a task that used to start with simply walking into an office.

Senior citizens are disproportionately affected. Many did not grow up with digital devices, may lack a personal email address or a smartphone capable of NFC-based ID verification, and can find multi-step authentication flows — such as registering for BundID, verifying identity via the electronic ID function of the German ID card, or generating a one-time PIN — confusing or intimidating. At the same time, family members who might otherwise help are often unavailable during working hours, and municipal support desks are frequently overwhelmed or accessible only by appointment. The result is a widening digital divide: seniors are pushed toward services they cannot independently use, while structured, low-cost, repeatable support to close that gap remains scarce.

Existing initiatives ranging from municipal walk-in help desks to intergenerational digital-mentoring formats address parts of this problem, but tend to be one-off, location bound, or dependent on ad-hoc volunteer availability, without shared structure, standardized guides, or support tooling for the volunteers themselves. Few, if any, directly combine trained volunteers with reusable, task-specific instructions and an AI assistant that helps the volunteer answer context-specific questions without taking over sensitive data entry. This gap — between the availability of well-meaning helpers and the availability of a structured, safe way for those helpers to be effective and consistent — is what BridgeBuilders sets out to close.

3. Our Solution: BridgeBuilders

BridgeBuilders is a web application that connects student volunteers with senior citizens for in-person digital-assistance appointments. Rather than replacing human contact with automation, the concept keeps a person — a prepared, trained student — at the centre of every interaction, and uses technology to make that student more effective, more consistent, and safer for the senior they are helping.

The service works as follows: a senior citizen (or an organization on their behalf) books an appointment with a student volunteer through the BridgeBuilders website. Ahead of and during the appointment, the student opens the relevant guide on their phone or tablet, walks the senior through the steps together, generates a large-print, take-home handout of that guide, and — if a question arises that the guide does not cover — consults the built-in AI assistant for a quick, guide-grounded clarification.

Two user groups are addressed directly, each with a distinct perspective on the service:

3.1 Student Side

For students, BridgeBuilders is designed to lower the barrier to volunteering rather than add administrative overhead. Instead of preparing ad-hoc for every appointment, a student opens the app, picks the guide that matches what the senior needs, and has a ready script, tips and what-if scenarios at hand. The dashboard tracks onboarding progress and volunteering hours, which can support applications for certificates of engagement. Motivation for participation was discussed within the team, including certificates of engagement and forms of practical benefit such as reduced-cost housing arrangements modelled on existing “living for help” (Wohnen für Hilfe) schemes; these mechanisms were explored as options during ideation rather than finalized as part of the prototype.

3.2 Senior Side

For seniors, the service replaces an anonymous phone hotline or written FAQ with a patient, face-to-face conversation at a scheduled appointment. Seniors keep a large-print, take-home handout for every guide they work through, so they can retrace the steps on their own afterwards, and the AI assistant is never exposed to the senior directly — it remains a support tool for the student, not a replacement for the personal interaction.

3.3 Future Plans

Beyond the initial four guides, the team envisions extending BridgeBuilders to cover the remaining topics in the four guide categories (Communication, Government Services, Smartphone Basics, and Security), integrating a lightweight appointment-booking calendar directly with partner organizations, and — pending a full legal review — piloting the service with a small cohort of student volunteers and senior participants (see Section 7).

4. Creation of Guides

A central design decision was to not leave the content of the assistance to chance. The team developed a structured Guide Library: step-by-step guides across four categories (Communication, Government Services, Smartphone Basics, and Security) each broken down into individual steps with expandable “what-if scenarios” for common problems. Four of these guides address the most frequent digital tasks identified during our research and are described below. [Bridge Builders Guidelines](#)

4.1 Creating a New Email Account

Many administrative processes assume that a citizen has a personal email address, yet a substantial share of seniors either do not have one or are unsure how to create or use one. This guide walks the student and senior through choosing an email provider, setting up a memorable and secure address, choosing a password the senior can manage, and confirming the account by sending and receiving a first test message.

4.2 Creating a BundID / eID

BundID is the central identity account used to access most German federal online services. This guide covers the most common two registration options basic access via username and password, verification via the elektronischer Identitätsnachweis (eID) function of the German ID card and explains, in plain language, what each option unlocks and what hardware (an NFC-capable smartphone or card reader) it requires. Particular attention is given to the eID PIN, which the guide reminds students never to write down or store on the senior’s behalf

4.3 Purchasing a Digital Train Ticket

The reduction of physical ticket offices forces seniors to use digital ticketing platforms that trigger anxiety. This guide structures the purchase of a ticket or a *Deutschlandticket* via the *DB Navigator* app. The student must point to the screen to guide the senior through the route fields ("Von"/"Nach") and calendar selections, ensuring the user types each letter individually. During the tariff phase, differences between express (ICE) and regional trains (RE/BRB) must be explained objectively, and discount profiles like a *BahnCard* should be applied via the tab.

The student must remain neutral regarding payment methods turning away the moment the screen opens to protect data privacy. To mitigate panic during cellular dead zones, the senior must be taught how to screenshot the resulting QR code. Finally, the app must be closed and reopened to validate the process through an independence test.

4.4 Booking a Bürgeramt Appointment

Registering a new address, renewing an ID card, or requesting official documents in Germany typically requires an appointment at the local citizens' registration office (Bürgeramt), most of which is now booked online. This guide walks through finding the correct local office, selecting the right service category, choosing an available slot, and saving the confirmation, including which documents to bring.

5. Legal & Safety Design

Volunteering to help seniors with digital tasks involves working in highly sensitive areas, such as medical appointments and tax portals. Instead of treating legal compliance as a dry checklist, we embraced it as our primary design challenge: *How do we empower student volunteers to be helpful, while guaranteeing that senior citizens' data remains absolutely secure?*

Through a critical reflection on potential risks, we translated complex European and German regulations into four user-centered design pillars.

5.1 Designing for Volunteer Safety

We recognized that students might feel anxious about accidentally mishandling a senior's personal information and devices. Guided by the statutory liability limits for gratuitous volunteer assistance (§ 675 BGB) and German data espionage laws (§ 202a StGB), we implemented our strict "Golden Rule". Volunteers act exclusively as verbal guides and are structurally prohibited from physically touching a senior's screen or typing on their behalf. This transparent rule shields students from liability and ensures that the senior is always in the driver's seat.

5.2 Zero Data Retention

To address privacy issues, we directly applied GDPR Article 5 (data minimization) and Article 9 (prohibitions on sensitive category data). Our software automatically blocks passwords or official ID numbers from ever being entered into the chat. Furthermore, to comply with GDPR Article 32 (security of processing), we built an automated 48-hour "self-destruct" lifecycle for any basic booking data (such as a first initial or an encrypted phone number). Exactly 48 hours after an appointment ends, all coordination logs are permanently erased from our servers.

5.3 Keeping AI Helpful, Not Risky

We wanted the power of AI without the legal gray areas. To consciously prevent our tool from entering the "High-Risk" territory defined by Annex III of the EU AI Act, we strictly limited the assistant's capabilities. Following Article 50 of the AI Act, we ensured every AI response is transparently labeled so users always know they are interacting with a machine. We also pressure-tested it against edge cases: if a user asks, "How do I log into someone else's tax account?", the assistant is programmed to firmly refuse and redirect to official guides.

5.4 Accessible for Everyone

We firmly believe that true compliance isn't just about protecting data; it's about accessibility. To meet the requirements of the European Accessibility Act (EAA) and the German BFSG, we integrated large text, high-contrast, and screen-reader support not as optional add-ons but as the very foundation of our interface.

5.5 Conclusion: Invisible Safeguards

Ultimately, our transparent work process allowed us to turn a heavy stack of legal requirements—Privacy Policies, AI Governance, and Data Secrecy mandates (§ 53 BDSG)—into invisible, automated safeguards. By doing so, the senior retains 100% control over their digital life, and the student volunteers within a completely safe, legally sound environment.

6. Prototype

To validate the concept beyond a purely conceptual level, the team implemented a functional prototype of the BridgeBuilders web application.

6.1 What the Prototype Demonstrates (Student Side)

The prototype implements the core end-to-end experience a student would go through during a real appointment. Table 1 summarizes the five core features implemented.

Feature	What it does
Home Dashboard	Daily greeting, onboarding progress (%), next scheduled workshop appointment, and quick-access tiles to the rest of the app.
Guide Library	Eight step-by-step guides across four categories (Communication, Government Services, Smartphone Basics, Security), each shown as an accordion with expandable, practical tips.
Handout Generator	Exports any guide as a print-optimized PDF or RTF file, in one of two enlarged font sizes (Large / Very Large) for seniors.
AI Assistant	Context-aware chat grounded in the guide content, with Markdown-formatted responses and clear boundaries for legal or social questions it should not answer directly.
Profile & Language	DE / EN toggle, volunteering-experience status, and hours tracking for the student.

Table 1. Core features implemented in the BridgeBuilders prototype for the students.

6.2 What the Prototype Demonstrates (ASZ / Administration Side)

The prototype also covers the ASZ or Administration side that manages guides, students, workshops and programme logistics. The student portal is optimised for mobile-first volunteer workflows, while the ASZ portal is a desktop-oriented management dashboard accessed at ASZ. The portal has closed-loop workflows. When ASZ staff publish a guide or workshop, it becomes visible in the student guide library or workshop booking flow. When students log hours or submit suggestions, those entries appear in the ASZ dashboard and affect the student's

experience status, which unlocks responsibilities including workshop supervision. Table 2 summarizes the core feature areas that implement these workflows.

Feature	Description
Overview Dashboard	KPI cards for pending hours, total logged hours, and upcoming workshop slots; one-click confirmation of recent unconfirmed entries
Appointment Management	Create individual and workshop slots (title, time, room, capacity, linked workshop plan); edit/delete workshops; consolidated booking table with student names and timestamps; server-side overbooking prevention.
Hour Review & Billing	Per-student overview (entry count, total minutes, last report) with detail drill-down; confirm individual entries or bulk-confirm all; CSV export; confirmed hours' drive student experience flags.
Content Management	CRUD for onboarding modules, guides, workshop plans, FAQ entries, and general PDF documents — all per locale (DE/EN); structured steps, pitfalls, optional PDF attachments, publish/draft toggles
Improvement Suggestions	Review student feedback on guides and workshop plans; filter by status and target type; add ASZ comments, update status, jump to the referenced content for editing

Table 1. Core features implemented in the BridgeBuilders prototype for the ASZ/Administration.

6.3 Technical Implementation

The ASZ portal shares the same foundation as the student side (React 18 + Vite frontend, FastAPI/Python backend, JWT auth, DE/EN i18n via React Context). The following phases cover ASZ-specific work on top of that base.

Phase 1: Role Routing & API Guards

- ASZ accounts created by admin only (role=asz); login via shared POST /auth/login, redirect to /asz
- Server-side require_roles(...) on all ASZ endpoints; client-side ProtectedRoute guard
- REST client layer (apiClient) with typed methods for slots, hours, guides, onboarding, FAQ, documents, and suggestions

Phase 2: Content & Onboarding Editor

- OnboardingEditor component: fixed module/quiz/agreement slot structure, editable title and body per locale
- Guide and workshop-plan forms with {title, locale, category, steps[], pitfalls[]} payload; validation before save
- FAQ editor with category, sort order, and publish flag; general-document upload by title and locale
- PDF storage on server filesystem (backend/data/uploads/); upload/delete via multipart endpoints

Phase 3: Operations & Workflows

- Workshop slot modal: datetime picker, room, capacity, supervisor slots, workshop-type selector, linked plan
- Hour confirmation flow: single confirm, confirm-all, and CSV export; overview aggregated by student
- Suggestion workflow: status transitions, ASZ comment field, deep-link to content editor (inhalte view)
- Scrollable DataTable components and KPI cards for desktop-first management UI

Phase 4: Demo

- Walkthrough recorded: Login → Overview (confirm hours) → Appointments (create slot) → Content (edit guide) → Suggestions (resolve feedback)

Technology Stack: React 18, Vite, FastAPI, Python, SQLite · JWT auth · OpenRouter (student chat only) · DE / EN

MVP scope notes: No payout logic, certificate generation, or university SSO. SQLite for pilot; production target is PostgreSQL with RLS. Individual appointment cancellation by students is not included.

GitLab Repository: <https://gitlab.lrz.de/000000003B9B678F/digitalrat> (Youssef has access)

7. Discussion & Outlook

The BridgeBuilders concept and prototype demonstrate that a lightweight, well-scoped combination of human volunteers, structured content, and a narrowly-scoped AI assistant can meaningfully lower the barrier for senior citizens to complete essential digital tasks — without requiring the senior to become independently digitally literate overnight, and without asking a volunteer to handle sensitive data on the senior's behalf.

7.1 Expected Impact on Students and Seniors

For seniors, the expected outcome is a lower anxiety, higher success rate for basic digital tasks. Instead of giving up halfway, they complete the process with a human present and leave with information. Repeated exposure to the same guided process is also intended to gradually build confidence and independence.

Students can expect a structured, low-effort way to get into civic volunteering. The format is clear, the hours and certificates are tracked, and the activity is easy to commit to alongside studies. The guardrails in Section 5 protect students from being asked to take on responsibilities that they should not be expected to take on as volunteers.

7.2 Limitations

Several questions remain. The prototype relies on demo accounts and has not yet been tested with real participants, so usability assumptions still require field validation. The legal-safeguards framework has not undergone formal legal review; before any real-world pilot, the team would seek input from a data-protection officer and monitor further guidance on AI systems used in non-governmental contexts. Recruiting and retaining a sufficient, reliable pool of student volunteers is likely to be the biggest practical obstacle to scaling beyond a single university or district.

7.3 Future Expectations and Roadmap

In alignment with our strategic vision, a three-phase roadmap has been developed to scale the impact of the BridgeBuilders project:

- **Short-Term (0–12 Months):** Pilot programs will be launched at selected Alten- und Service-Zentren (ASZ) to gather feedback and refine training materials.
- **Medium-Term (1–3 Years):** Following the pilot phase, the service will be expanded across Munich. Concurrently, the volunteer pool will be enlarged through enhanced recruitment strategies, and partnerships with local universities will be strengthened to ensure long-term sustainability.

- Long-Term (3–5+ Years): In this final stage, the model will be integrated into university seminars. Furthermore, expansion to nearby cities will be pursued, and the service framework will be adapted to better accommodate the needs of people with disabilities.

Author Contribution

Name	Chapters	Name	Chapters
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Victor Tutu-Biaw	5 (50%), 7 (50%)	Ali Furkan Kiraci	5 (50%), 7 (50%)

All of us: Chapters 1, 2, 3

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